K.R. MANGALAM UNIVERSITY

Database Management Lab File

(ENCS205)

Bachelor of Technology

in

Computer Science and Engineering

Submitted BY

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B. Tech CSE (Core) – 4th Semester

Section: A

Submitted To:

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LAB SHEET 4

OBJECTIVE

Design and execute SQL commands to create users, manage their access privileges, and remove users in a database system. This assignment focuses on database security and access control mechanisms.

Lab Assignment: SQL User Management & Access Control

Title:

Implementation of **User Creation, Deletion, and Privilege Management in SQL**

Problem Statement

Design and execute SQL commands to **create users, manage their access privileges, and remove users** in a database system. This assignment focuses on **database security and access control mechanisms**.

Tasks

Part A: User Creation

1. Create a new user named `user1` with a password.
2. Create another user `user2` with limited access.

Part B: Granting Privileges

3. Grant **SELECT and INSERT** privileges on a table `Student` to `user1`.
4. Grant **ALL privileges** on a table `Course` to `user2`.
5. Provide **WITH GRANT OPTION** to `user1` and observe its effect.

Part C: Revoking Privileges

6. Revoke **INSERT permission** from `user1` on the `Student` table.
7. Revoke all privileges from `user2` on the `Course` table.

Part D: User Deletion

8. Drop/Delete user `user2` from the database.
9. Attempt to login using deleted credentials and observe the result.

Part E: Additional Commands

10. Display all users in the system.
11. Show privileges assigned to a specific user.

CODE:

```
CREATE DATABASE IF NOT EXISTS college;

USE college;


CREATE TABLE IF NOT EXISTS Student (

    student_id INT PRIMARY KEY AUTO_INCREMENT,

    name VARCHAR(50),

    age INT,

    course VARCHAR(50)

);


CREATE TABLE IF NOT EXISTS Course (

    course_id INT PRIMARY KEY AUTO_INCREMENT,

    course_name VARCHAR(50),

    duration VARCHAR(20)

);


CREATE USER IF NOT EXISTS 'user1'@'localhost' IDENTIFIED BY 'password123';

CREATE USER IF NOT EXISTS 'user2'@'localhost' IDENTIFIED BY 'password123';
```

```
GRANT SELECT, INSERT ON college.Student TO 'user1'@'localhost';
GRANT ALL PRIVILEGES ON college.Course TO 'user2'@'localhost';

GRANT SELECT, INSERT ON college.Student
TO 'user1'@'localhost'
WITH GRANT OPTION;

SHOW GRANTS FOR 'user1'@'localhost';
SHOW GRANTS FOR 'user2'@'localhost';

REVOKE INSERT ON college.Student FROM 'user1'@'localhost';
REVOKE ALL PRIVILEGES ON college.Course FROM 'user2'@'localhost';

SHOW GRANTS FOR 'user1'@'localhost';
SHOW GRANTS FOR 'user2'@'localhost';

DROP USER IF EXISTS 'user2'@'localhost';

SELECT User, Host FROM mysql.user;

SHOW GRANTS FOR 'user1'@'localhost';

FLUSH PRIVILEGES;
```

OUTPUT:

Result Grid | Filter Rows: | Expo

Grants for user1@localhost
GRANT USAGE ON *.* TO `user1`@`localhost`
GRANT SELECT, INSERT ON `college`.`student...

Result Grid | Filter Rows: | Export

Grants for user2@localhost
GRANT USAGE ON *.* TO `user2`@`localhost`
GRANT ALL PRIVILEGES ON `college`.`course`...

Result Grid | Filter Rows: | Expor

Grants for user1@localhost
GRANT USAGE ON *.* TO `user1`@`localhost`
GRANT SELECT ON `college`.`student` TO `us...

Result Grid | Filter Rows: | Export

Grants for user2@localhost
GRANT USAGE ON *.* TO `user2`@`localhost`

Result Grid | Filter Rows:

User	Host
dept_user	localhost
dept_user2	localhost
mysql.infoschema	localhost
mysql.session	localhost
mysql.sys	localhost
root	localhost
user1	localhost
NULL	NULL

Result Grid | Filter Rows: | Export:

	Grants for user1@localhost
▶	GRANT USAGE ON *.* TO `user1`@`localhost`
	GRANT SELECT ON `college`.`student` TO `us...